

FRANKLIN, VA, USA

WMO: 723083

Lat: 36.698N

Lon: 76.903W

Elev: 13

StdP: 101.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.1	-16.3	0.9	-4.2	-13.0	1.2	-1.6	9.4	13.6	8.4	11.5	1.4	350	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	36.0	25.5	33.9	24.9	32.7	24.5	27.0	32.7	26.3	31.8	25.7	30.9	3.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.3	29.5	24.9	19.9	28.6	24.0	18.9	28.0	84.3	33.7	81.7	32.0	78.9	31.3	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	5.7	-12.2	37.4	4.3	2.3	-15.3	39.0	-17.9	40.4	-20.3	41.7	-23.5	43.4		
(5)			-12.7	28.2	4.0	0.9	-15.6	28.9	-17.9	29.4	-20.1	29.9	-23.0	30.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	5.0	6.6	10.1	15.6	20.2	24.6	26.6	25.8	22.6	16.5	10.6	7.0	(6)
(7)		DBStd	8.88	6.14	5.45	5.50	4.93	4.17	3.33	2.65	2.59	3.25	4.57	4.90	5.21	(7)
(8)		HDD10.0	557	179	123	68	8	0	0	0	0	0	4	51	124	(8)
(9)		HDD18.3	1833	414	331	260	112	30	2	0	0	6	91	237	352	(9)
(10)		CDD10.0	2747	24	27	72	176	318	439	516	489	379	207	68	32	(10)
(11)		CDD18.3	980	1	1	6	29	89	191	257	231	135	34	4	2	(11)
(12)		CDH23.3	9578	2	8	63	308	871	1943	2717	2250	1097	284	28	8	(12)
(13)		CDH26.7	4046	0	0	11	86	311	873	1299	1003	389	72	2	0	(13)
(14)	Wind	WSAvg	2.3	2.5	2.6	2.9	2.9	2.5	2.2	2.0	1.8	1.9	2.0	2.1	(14)	
(15)	Precipitation	PrecAvg	1225	91	77	101	79	102	105	139	124	136	95	85	87	(15)
(16)		PrecMax	1602	163	162	240	172	174	255	283	332	485	198	215	183	(16)
(17)		PrecMin	877	24	23	12	21	32	41	39	36	14	0	18	32	(17)
(18)		PrecStd	196	37	36	48	37	37	55	62	71	105	52	54	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	24.1	27.8	32.1	33.9	37.3	38.0	37.1	33.9	31.4	26.3	24.0	(19)
(20)			MCWB	17.5	17.4	18.5	20.4	23.2	25.2	25.8	26.6	24.1	22.7	19.2	18.1	(20)
(21)		2%	DB	20.5	21.4	25.0	28.9	32.1	34.7	36.1	34.8	32.3	28.5	23.7	21.2	(21)
(22)			MCWB	16.8	15.4	16.8	19.4	22.5	24.6	25.8	25.5	24.0	21.8	17.9	18.0	(22)
(23)		5%	DB	17.7	18.8	22.5	27.3	30.0	32.8	33.9	33.0	30.9	26.7	22.0	18.6	(23)
(24)			MCWB	14.7	14.3	15.2	18.7	21.9	24.3	25.5	24.9	23.5	20.3	17.5	16.3	(24)
(25)		10%	DB	15.1	16.4	19.9	24.4	27.9	32.0	32.6	32.1	28.9	24.1	19.0	16.3	(25)
(26)			MCWB	11.8	11.7	14.5	17.3	20.8	23.9	25.0	24.8	22.6	19.4	15.6	13.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	19.0	20.0	22.5	25.1	27.0	28.0	28.0	26.5	24.6	21.3	20.5	(27)
(28)			MCDWB	21.4	22.4	25.3	27.8	31.8	33.4	34.0	34.7	29.3	27.7	24.0	22.5	(28)
(29)		2%	WB	17.4	17.3	18.6	21.1	23.8	25.8	27.1	26.9	25.5	23.3	19.5	18.5	(29)
(30)			MCDWB	19.4	19.5	22.9	26.7	29.2	32.2	33.0	32.6	29.3	26.4	22.3	20.5	(30)
(31)		5%	WB	15.3	15.3	17.3	19.9	22.9	25.2	26.4	26.1	24.7	22.0	18.0	16.7	(31)
(32)			MCDWB	17.4	18.6	20.4	25.1	27.7	31.3	32.4	31.2	28.5	25.5	20.5	18.3	(32)
(33)		10%	WB	12.7	12.6	15.5	18.6	21.9	24.6	25.8	25.4	23.9	20.5	16.5	14.2	(33)
(34)			MCDWB	14.7	15.4	19.5	23.0	26.7	30.0	31.3	30.0	27.4	23.5	19.0	16.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.8	12.5	13.1	12.2	11.6	11.1	11.0	11.2	12.5	12.5	11.2	(35)
(36)			MCDBR	13.6	14.9	15.4	15.1	13.9	13.0	12.5	12.4	12.8	14.0	14.5	13.0	(36)
(37)			MCWBR	10.5	10.4	8.9	7.3	6.3	5.2	4.9	5.0	5.7	7.5	9.5	9.9	(37)
(38)		5% WB	MCDWB	12.5	13.8	12.9	12.9	11.8	11.4	11.1	10.8	10.2	10.9	11.7	11.7	(38)
(39)			MCWBR	10.8	11.0	9.3	7.4	5.9	5.0	4.8	4.8	5.2	6.4	8.8	9.9	(39)
(40)																
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.319	0.347	0.391	0.437	0.491	0.518	0.511	0.406	0.357	0.322	0.308	(40)	
(41)		taud	2.523	2.480	2.429	2.316	2.218	2.116	2.061	2.065	2.355	2.481	2.529	2.549	(41)	
(42)		Ebn at Noon	895	920	919	890	851	802	777	772	846	864	864	867	(42)	
(43)		Edh at Noon	79	93	107	127	142	157	165	161	114	92	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.41	3.14	4.15	5.28	5.92	6.35	6.03	5.44	4.44	3.55	2.70	2.13	(44)	
(45)		RadStd	0.20	0.35	0.33	0.40	0.49	0.40	0.30	0.30	0.37	0.43	0.22	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+111

Nomenclature: See separate page